

Implementation Report: EEG Noise-Removal Pipeline

LP Praktikum Informatik 1

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Implementation

EEG noise-removal pipeline

Focus: exact dataset justification → preprocessing → synthetic artifacts → bad-channel

detection/interpolation → ICA blink removal

Dataset Used: S001R01.edf

Exact data description and justification

Source and exact experiment

PhysioNet EEG Motor Movement/Imagery Dataset (EEGMMIDB)

File: S001R01.edf → Subject 1, Run 1

Run 1 = baseline eyes-open recording

64 EEG channels, 160 Hz sampling rate, ~61 seconds duration

Why this data?

Public, small, MNE-compatible, realistic multichannel EEG

Suitable for demonstrating preprocessing steps without task-classification complexity

Important limitation

No labeled EOG/EMG/bad-channel artifacts; original sample mainly supports line-noise/basic filtering demonstration

Solution to limitation: controlled synthetic artifacts were added for blink, muscle noise, flat channel, and noisy channel demonstrations.

Pipeline

Parts 4–12 of the notebook

Global bad-channel metrics

Filter + notch + CAR

Raw inspection

ICA blink removal

Persistent bad channels

Synthetic artifacts

Epoch-wise bad detection

Interpolation

Raw Data Inspection

Original EEG before preprocessing

Time-domain plot

Shows selected EEG channels over time.

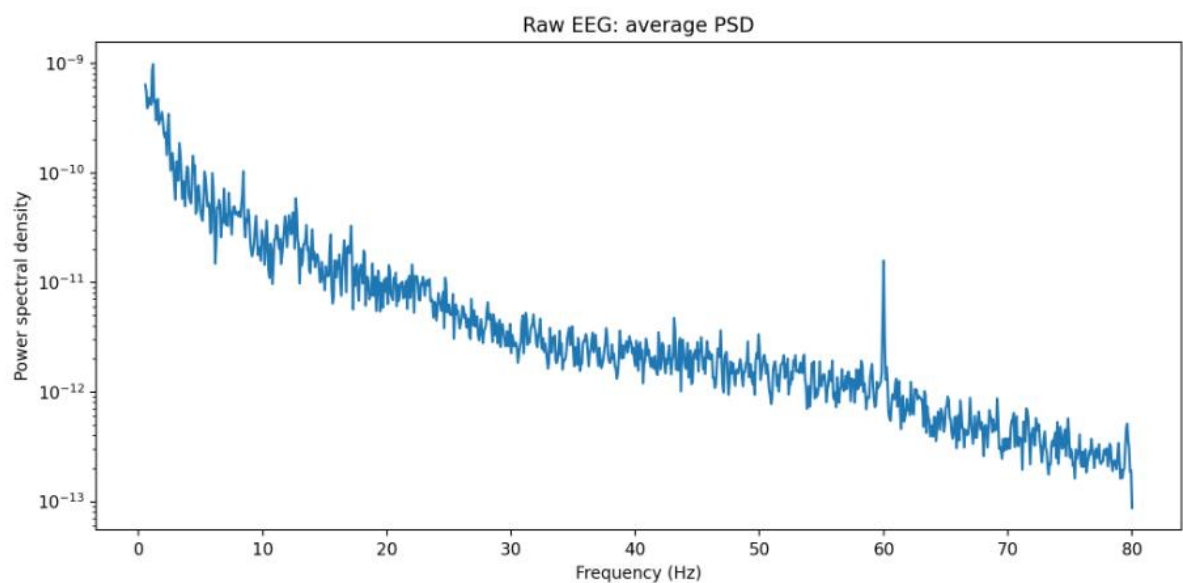
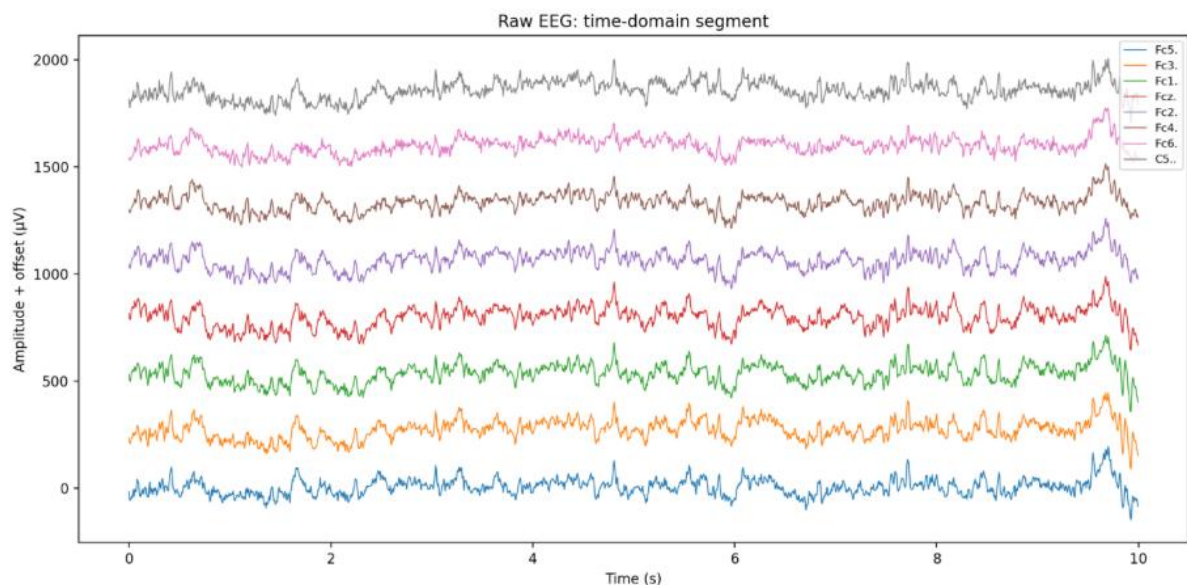
“Amplitude + offset” means channels are vertically shifted only for readability.

PSD plot

Shows power distribution over frequencies.

Raw PSD helps identify line noise and high-frequency components before filtering.

Takeaway: raw inspection justifies the next preprocessing steps and shows why both time-domain and frequency-domain views are useful.



Basic Preprocessing

1–40 Hz band-pass + 50 Hz notch + common average reference

Operations applied

1 Hz high-pass: reduces slow drift / baseline movement.

40 Hz low-pass: suppresses high-frequency noise and muscle-like activity.

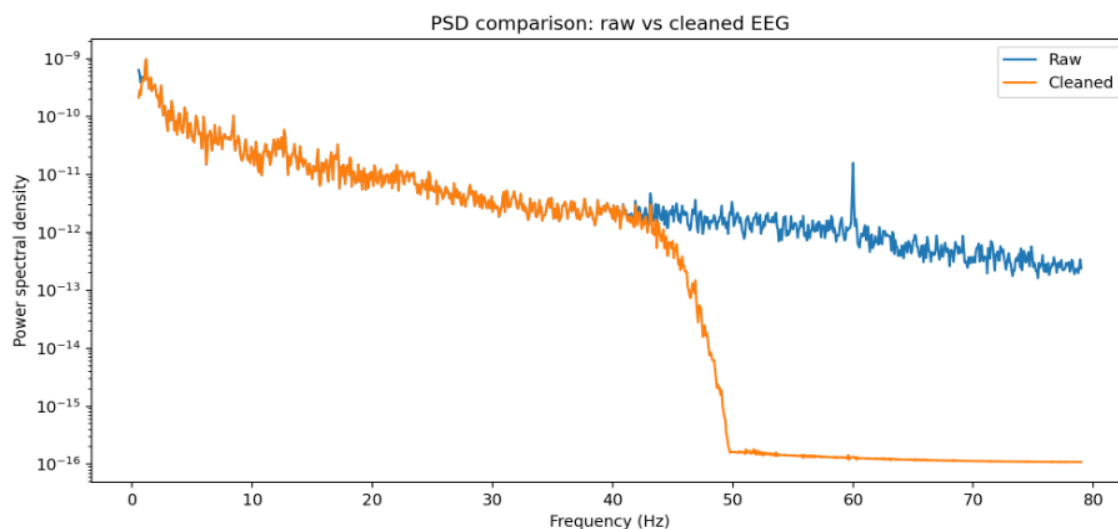
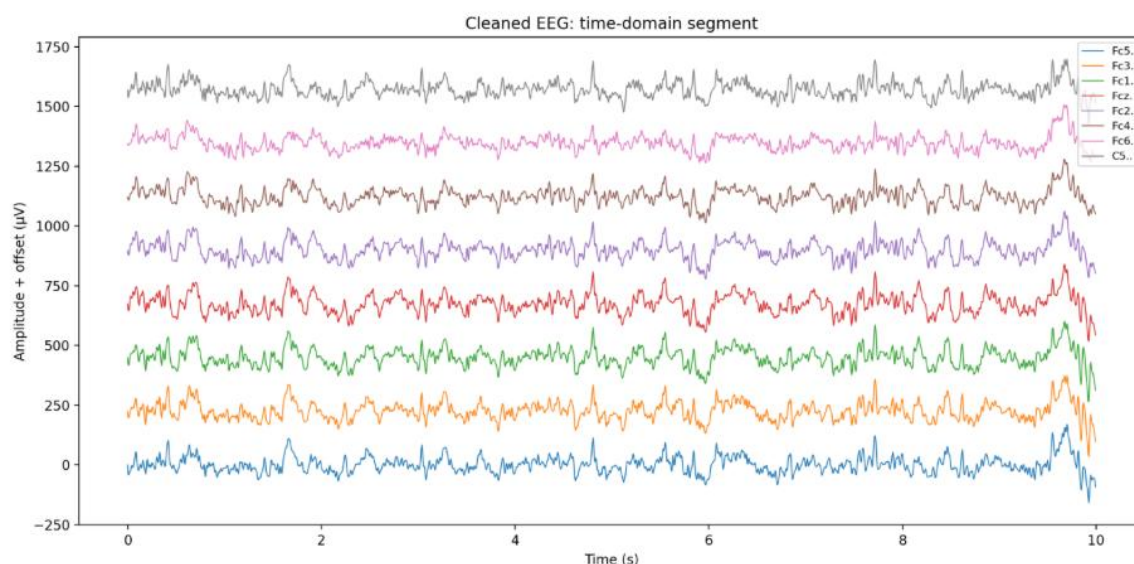
50 Hz notch: included as standard power-line-noise step.

Common average reference: subtracts average EEG activity from all channels.

Observation

PSD shows strong attenuation above the 40 Hz low-pass region.

Note: The dataset shows a visible peak around 60 Hz. With a 40 Hz low-pass, this region is already strongly attenuated; a 60 Hz notch would be more targeted if preserving >40 Hz frequencies was needed.



Synthetic Artifact Demonstration

Controlled artifacts added because the original file has no labeled artifact examples

Why synthetic artifacts?

Original S001R01.edf is baseline eyes-open and not labeled for EOG/EMG/bad-channel examples.

Synthetic artifacts allow controlled before/after demonstration.

Added artifacts

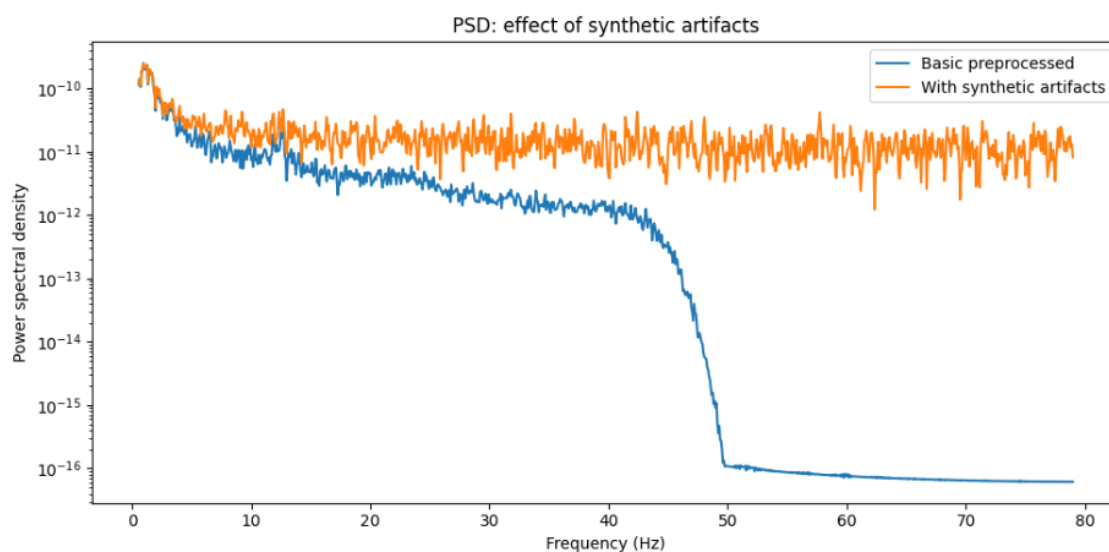
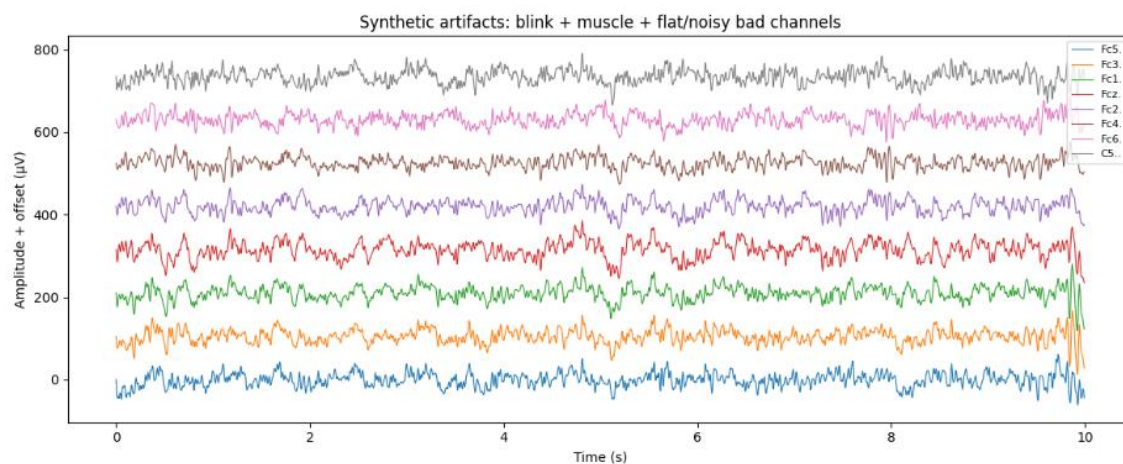
Blink-like slow wave → frontal channels Fp1, Fpz, Fp2.

Muscle-like high-frequency noise → temporal channels T7, T8, FT7, FT8.

Flat channel → Pz.

Noisy channel → O1.

Interpretation: synthetic artifacts increase signal power and create controlled examples for detection, interpolation, and ICA.



Global Bad-Channel Detection Metrics

Detect suspicious channels across the whole recording

Metrics per channel

Standard deviation: high = noisy; very low = flat.

Peak-to-peak amplitude: high = abnormal large swings.

Mean correlation with other channels: low = unusual behavior.

Robust z-score = $0.6745 \times (\text{value} - \text{median}) / \text{MAD}$.

Example output interpretation

Pz: flat synthetic channel → std = 0, peak-to-peak = 0.

O1: noisy synthetic channel → very high STD and peak-to-peak.

Fp1/Fp2/AF8: suspicious due to blink-like high-amplitude frontal activity.

Important distinction: “suspicious” does not always mean broken electrode. Blink-contaminated frontal channels should usually be handled by ICA, not interpolation.

Epoch-Wise Bad-Channel / Bad-Epoch Detection

Automatically tag channel-epoch pairs and full bad epochs

Method

Continuous EEG is split into 2-second epochs.

For each channel in each epoch: STD, peak-to-peak, and correlation with median signal are computed.

A channel is bad in an epoch if it is flat, noisy, high-amplitude, or poorly correlated.

An entire epoch is bad if >20% of channels are bad.

Observed output

30 epochs from ~61 seconds of EEG.

109 bad channel-epoch pairs.

Bad epochs: none → artifacts are local channel problems, not whole-recording failures.

Visualizing Epoch-Wise Detection

Heatmap of bad channel flags over time

How to read the heatmap

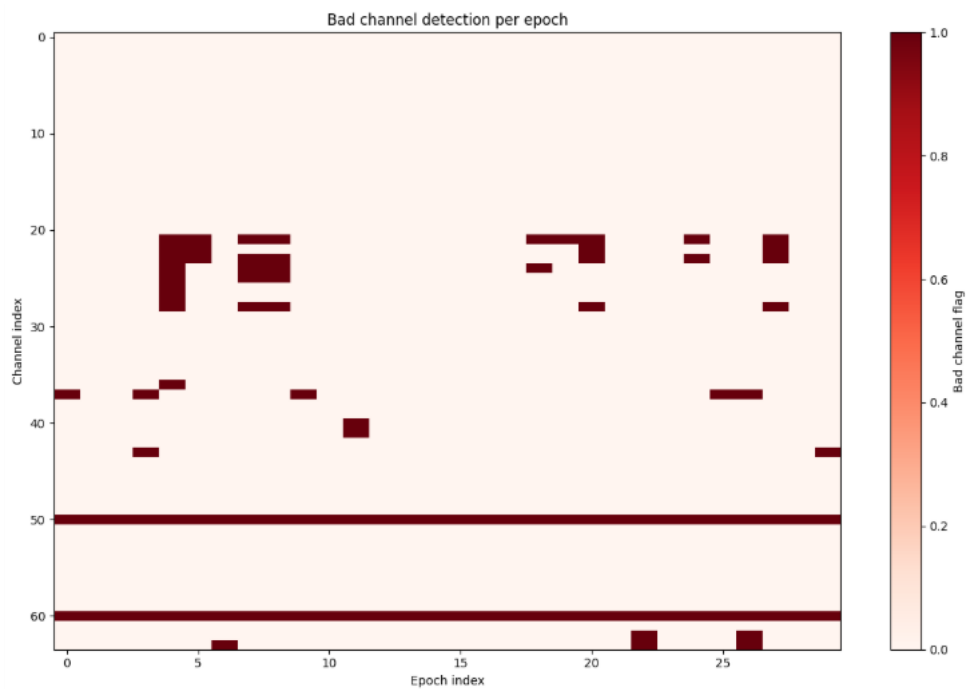
x-axis = epoch index; y-axis = channel index.

Red = channel marked bad in that epoch; light = channel normal.

Horizontal red lines = persistent bad channels.

Scattered red blocks = temporary channel-specific artifacts.

No full vertical red bands = no full bad epochs.



Persistent Bad Channels

Deciding which channels are bad often enough to interpolate

Definition

A persistent bad channel is bad in more than 30% of epochs.

With 30 epochs, this means bad in more than ~9 epochs.

Why this step matters

Prevents interpolating a good channel because of one temporary artifact.

Separates temporary artifacts from structural/persistent channel problems.

Expected result in the synthetic example

Pz: flat channel → persistent.

O1: noisy channel → persistent.

Blink-affected frontal channels should usually go to ICA, not interpolation.

Bad-Channel Interpolation

Reconstruct structural/persistent bad channels from neighboring electrodes

Interpolation idea

Bad channel signal is estimated from neighboring good electrodes.

Requires montage/electrode positions; channel names were cleaned to match standard_1020 montage.

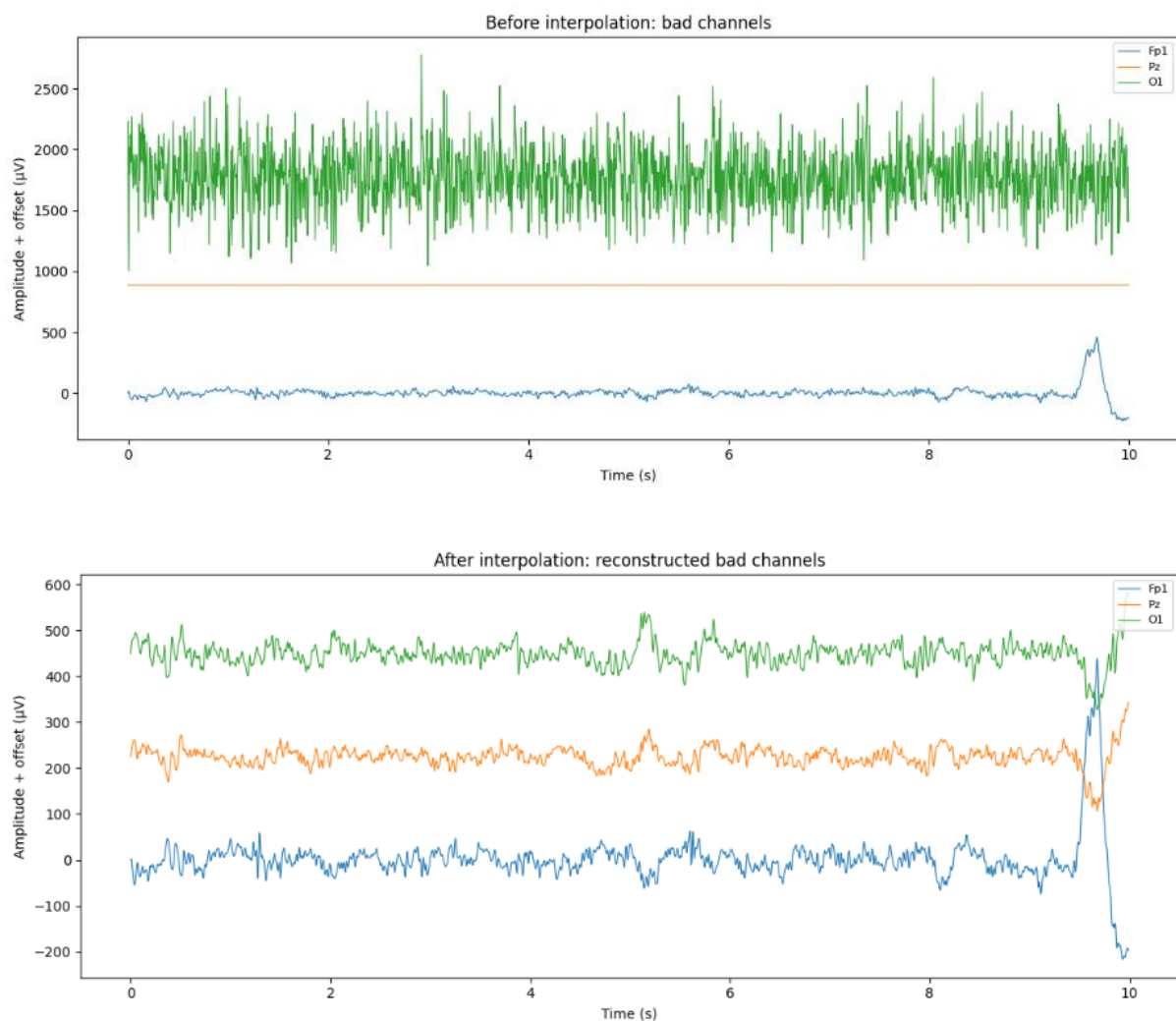
Example

Before: Pz is flat, O1 is extremely noisy, Fp1 has blink-like high amplitude.

After: bad channels are reconstructed and MNE bad list is reset.

Caveat

Pz/O1 are good interpolation examples; Fp1 is more artifact-contaminated and better handled with ICA in a real pipeline.



ICA Blink-Removal Example

Component-based removal of a synthetic frontal blink

How ICA is used

Add synthetic blink to Fp1, Fpz, Fp2.

Run ICA on multichannel EEG.

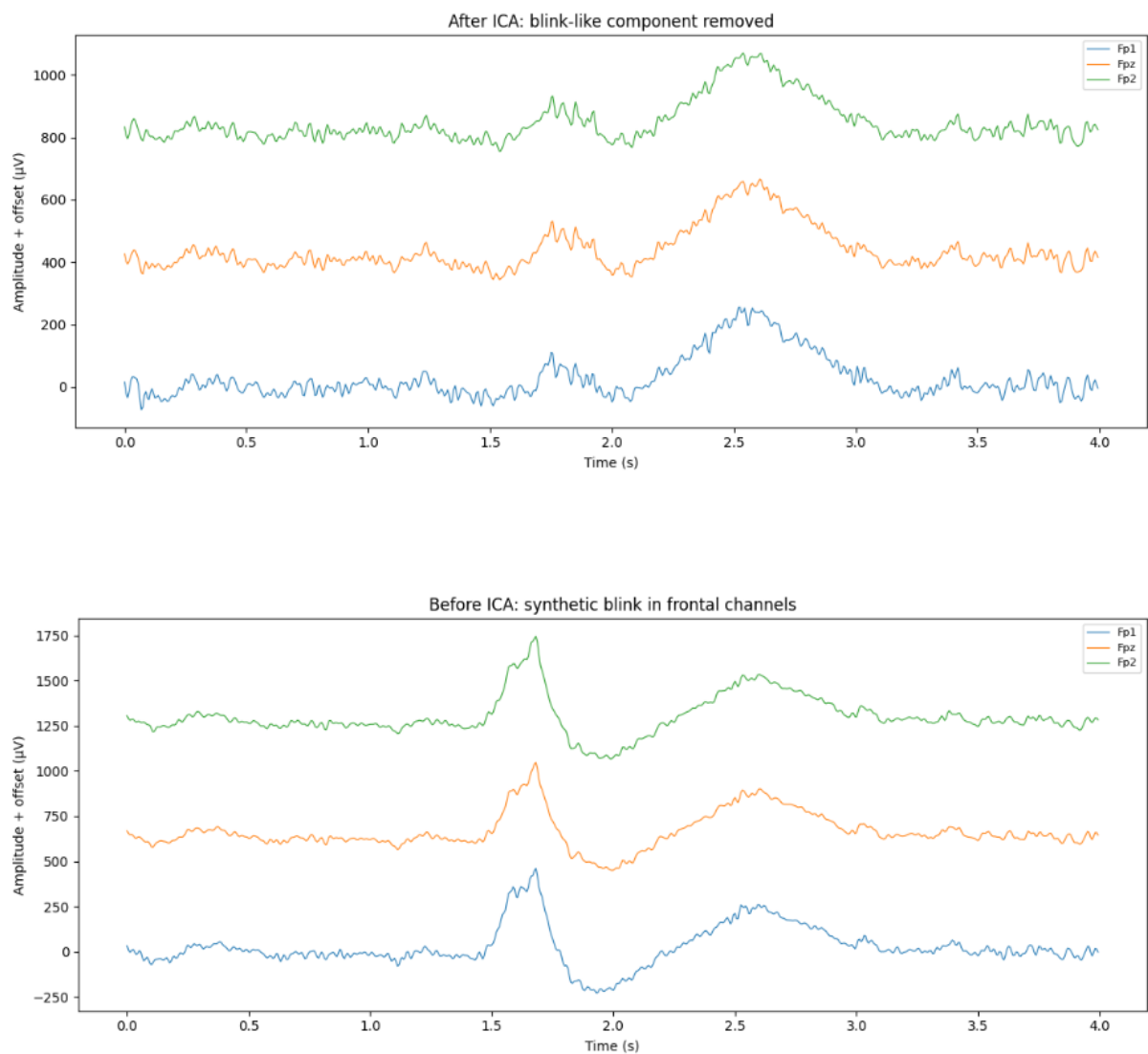
Select component most correlated with the frontal blink signal.

Remove that component and reconstruct EEG.

Interpretation

Blink-like ICA component should show a strong frontal topography.

After removal, the large frontal blink deflection is reduced.



Summary

2- Preprocessing / Demonstration Pipeline

1- Dataset

Raw EEG + PSD inspection

1–40 Hz band-pass + 50 Hz notch

Common Average Reference

Synthetic blink / muscle / bad-channel examples

Global channel-quality metrics

Epoch-wise bad-channel detection

Persistent bad-channel selection

Bad-channel interpolation

ICA blink-component removal

S001R01.edf

PhysioNet EEGMMIDB

64 EEG channels, 160 Hz, ~61 s

Baseline eyes-open recording

3- Main Outputs

Before/after time-domain plots

Before/after PSD plots

Bad-channel epoch heatmap

Interpolation comparison

ICA before/after comparison

EEG preprocessing: inspect → filter / re-reference → detect → interpolate / remove → compare

The GitHub repository link consisting of the regarding mentioned files is:

<https://github.com/Soroooush/LP-1>